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But $\rho = \frac{M}{V}$

Therefore,

$$v = \sqrt{\frac{\gamma PV}{M}}$$

Using equation (11.26)

$$v = \sqrt{\frac{\gamma RT}{M}}$$

Where, R is the universal gas constant and M is the molecular mass of the gas. The speed of sound in nitrogen gas at 17°C is

$$v_N = \sqrt{\frac{\gamma R(273K + 17K)}{M_N}} = \sqrt{\frac{\gamma R(290K)}{M_N}} \quad (1)$$

Similarly, the speed of sound in oxygen gas at t in K is

$$v_0 = \sqrt{\frac{\gamma R(273K + t)}{M_0}} \quad (2)$$

Given that the value of γ is same for both the gases, the two speeds must be equal. Hence, equating equation (1) and (2), we get

$$v_O = v_N$$

$$\sqrt{\frac{\gamma R(273 + t)}{M_0}} = \sqrt{\frac{\gamma R(290)}{M_N}}$$

Squaring on both sides and cancelling γR term and rearranging, we get

$$\frac{M_0}{M_N} = \frac{273 + t}{290} \quad (3)$$

Since the densities of oxygen and nitrogen is 16:14,

$$\frac{\rho_0}{\rho_N} = \frac{16}{14} \quad (4)$$

$$\frac{\rho_0}{\rho_N} = \frac{\frac{M_O}{V}}{\frac{M_N}{V}} = \frac{M_0}{M_N} \Rightarrow \frac{M_0}{M_N} = \frac{16}{14} \quad (5)$$

Substituting equation (5) in equation (3), we get

$$\frac{273 + t}{290} = \frac{16}{14} \Rightarrow 3822 + 14t = 4640$$

$$\Rightarrow t = 58.4\,\text{K}$$

11.5 REFLECTION OF SOUND WAVES

When sound wave passes from one medium to another medium, the following things can happen

(a) **Reflection of sound:** If the medium is highly dense (highly rigid), the sound can be reflected completely (bounced back) to the original medium.

(b) **Refraction of sound:** When the sound waves propagate from one medium to another medium such that there can be some energy loss due to absorption by the second medium.

In this section, we will consider only the reflection of sound waves in a medium when it experiences a harder surface. Similar to light, sound can also obey the *laws of reflection*, which states that

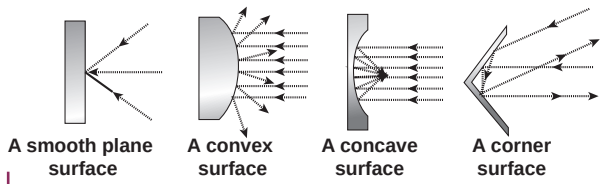


Figure 11.17 Reflection of sound in different surfaces